

Why polar excitons stay sharp: parity protection of the center-of-mass recoil channel in exciton-phonon scattering

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In polar semiconductors the Fröhlich interaction is the dominant electron-phonon coupling, yet excitonic resonances in materials such as halide perovskites remain anomalously sharp. We show that standard frozen-center-of-mass treatments of the exciton-phonon problem miss the decisive kinematic degree of freedom: restoring the exact center-of-mass (COM) recoil reveals a universally open, parameter-free $1s \rightarrow 1s$ absorption channel at recoil momentum $q_* = \sqrt{2M_{\text{ex}}\hbar\omega_{\text{LO}}}/\hbar$, whose rate scales as $N_{\text{LO}}(T)$. We prove that this recoil channel is controlled by destructive electron-hole interference: the recoil linewidth vanishes with the mass asymmetry as $\gamma_{\text{LO}}^{\text{recoil}} \propto \mathcal{F}_{1s,1s}(q_*)^2$, and the elastic dressing obeys the exact suppression law $S_X/S_{\text{ind}} = \eta^2(6 - \eta^2)/5$ within the hydrogenic Fröhlich model. The theory establishes a hierarchy of scattering regimes. In mass-asymmetric materials (GaAs, $\eta = -0.74$) the recoil channel is active ($\gamma_{\text{LO}}^{\text{recoil}} = 2.2$ meV); in mass-symmetric materials (FAPbI₃, $\eta = 0$; MAPbI₃, $\eta = -0.11$) it is killed by interference (0.00 and 0.12 meV), showing that the observed 27–40 meV perovskite linewidths cannot be accounted for by COM recoil and therefore require internal-state-changing and other inelastic channels, of which the constructive, η -robust $1s \rightarrow np$ resonance is the leading candidate within the present model. The Fröhlich constant α alone is therefore insufficient as a figure of merit: after projection onto the correlated exciton, the controlling parameters are η , q_*a_X , and the Rydberg detuning.

I. INTRODUCTION

The interaction of excitons with polar optical phonons governs the coherence of optical excitations in semiconductors, from conventional III–V compounds to halide perovskites and monolayer chalcogenides [1, 4, 5]. The standard description treats the electron and hole as independent Fröhlich polarons [2], an approach that predicts strong lattice dressing whenever the Fröhlich constant α is large. Yet materials with $\alpha \sim 2$ –3, such as lead-halide perovskites, exhibit sharp zero-phonon lines and modest homogeneous broadening [6, 7], a discrepancy that has motivated extensive phenomenological fitting of temperature-dependent linewidths [5, 7].

The independent-polaron picture neglects that a phonon couples to the *combined* charge density of a correlated electron-hole pair. Many-body perturbation theory (MBPT) in the *GW*–Bethe–Salpeter (BSE) framework [8–14] provides the natural language for such correlated excitations, and recent work has begun to construct exciton-phonon self-energies beyond independent quasiparticles. In Ref. [15] we derived the exact multipole structure of the excitonic Fröhlich vertex and established a parity-controlled interference selection rule; in Ref. [16] we quantified, against exact diagonalization, when the underlying *GW* description itself may be trusted.

Here we derive the observable spectral consequences of that selection rule using only analytically tractable models, and establish three results. (i) The internal-state-preserving vertex is infrared-safe and the Huang–Rhys factor obeys an exact suppression law, Eq. (6). (ii)

The inelastic vertex is finite, dipolar, and robust against mass asymmetry. (iii) The homogeneous linewidth is kinematically gated: the recoil coefficient $\gamma_{\text{LO}}^{\text{recoil}}$ is a parameter-free number fixed by (η, q_*a_X) , while the internal-state-changing channels are gated by the detuning $|\Delta E_{2p} - \hbar\omega_{\text{LO}}|$. Together these results explain why polar excitons stay sharp and identify the design parameters that control exciton-phonon scattering.

II. THEORY

A. Exciton-projected Fröhlich vertex

We describe the relative motion of the exciton by hydrogenic envelope functions $\psi_{nlm}(\mathbf{r})$ with Bohr radius a_X , and couple it to a dispersionless LO mode through the Fröhlich vertex $M(q) = i[2\pi e^2\hbar\omega_{\text{LO}}/(Vq^2)]^{1/2}(\varepsilon_\infty^{-1} - \varepsilon_s^{-1})^{1/2}$ [1]. Projecting onto the relative coordinate, the electron and hole sample momenta $f_e\mathbf{q}$ and $-f_h\mathbf{q}$, where $f_{e,h} = m_{h,e}/(m_e + m_h)$ are the mass fractions. The excitonic vertex is

$$G_{SS'}(q) = M(q) \langle S' | e^{if_e\mathbf{q}\cdot\mathbf{r}} - e^{-if_h\mathbf{q}\cdot\mathbf{r}} | S \rangle \equiv M(q) \mathcal{F}_{SS'}(q). \quad (1)$$

All physics below follows from the excitonic form factors $\mathcal{F}_{SS'}(q)$, both their small- q multipole structure and their finite- q values at the recoil momentum q_* .

B. Internal-state-preserving channel: exact parity protection

We first consider scattering processes that preserve the internal excitonic quantum numbers ($1s \rightarrow 1s$) but in-

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volve a change in the center-of-mass (COM) momentum. The excitonic form factor is

$$\mathcal{F}_{1s,1s}(q) = \langle 1s | e^{if_e \mathbf{q} \cdot \mathbf{r}} - e^{-if_h \mathbf{q} \cdot \mathbf{r}} | 1s \rangle. \quad (2)$$

For a mass-symmetric exciton ($m_e = m_h \Rightarrow \eta = 0$) [20], the mass fractions are exactly $f_e = f_h = 1/2$. Eq. (2) then reduces to

$$\mathcal{F}_{1s,1s}(q, \eta = 0) = 2i \langle 1s | \sin(\mathbf{q} \cdot \mathbf{r}/2) | 1s \rangle. \quad (3)$$

Because the hydrogenic $1s$ envelope is purely real and spherically symmetric (even parity), and $\sin(\mathbf{q} \cdot \mathbf{r}/2)$ is an odd function of $\mathbf{r}$, the integrand is strictly odd. Consequently, $\mathcal{F}_{1s,1s}(q, \eta = 0) = 0$ *identically for all magnitudes of q* . This is a much stronger result than the small- q expansion: the internal-state-preserving coupling to the macroscopic polar field is exactly forbidden by parity for any mass-symmetric exciton, regardless of the recoil momentum. For later use we record the closed-form hydrogenic result $\langle e^{i\mathbf{k} \cdot \mathbf{r}} \rangle_{1s} = (1 + k^2 a_X^2/4)^{-2}$, whence

$$\mathcal{F}_{1s,1s}(q) = (1 + f_e^2 x^2/4)^{-2} - (1 + f_h^2 x^2/4)^{-2}, \quad x \equiv qa_X, \quad (4)$$

which vanishes identically for $f_e = f_h$ (all q). The exact symbolic multipole residual of Eq. (4) is

$$\mathcal{F}_{1s,1s}(q) = \frac{\eta}{2} (qa_X)^2 - \frac{3\eta(1 + \eta^2)}{32} (qa_X)^4 + \mathcal{O}(q^6), \quad (5)$$

so that the monopole and dipole vanish identically and the leading quadrupole term removes the $1/q$ Fröhlich divergence.

For mass-asymmetric excitons ($\eta \neq 0$), the cancellation is incomplete. The Huang–Rhys [3] factor $S_X = \sum_q |G_{1s,1s}(q)|^2 / (\hbar\omega_{\text{LO}})^2$ is infrared-finite and, evaluating the integral in closed form (Sec. III), obeys the exact suppression law

$$S_X = R(\eta) S_{\text{ind}}, \quad R(\eta) = \frac{\eta^2(6 - \eta^2)}{5}, \quad (6)$$

where $S_{\text{ind}} = S_e + S_h$ is the independent-polaron result [2]. $R(\eta)$ is exact for all η : it reduces to $C\eta^2$ with

$C = 6/5$ at small η and saturates at $R(1) = 1$, the limit in which a heavy point-like carrier cannot be canceled.

Relation to other work. This work extends the leading-order parity selection rule of Ref. [15] to an exact all- q identity, Eq. (4); Ref. [15] focuses on the inelastic multipole expansion, the dressing phase diagram, and the 2D extensions.

C. On-shell linewidths: COM recoil and inelastic resonance

A common approximation freezes the exciton COM at $\mathbf{K} = 0$, artificially closing the $1s \rightarrow 1s$ absorption channel for a dispersionless LO mode. Restoring the COM dispersion $E_X(\mathbf{K}) = E_{1s} + \hbar^2 K^2 / 2M_{\text{ex}}$ reveals a universally open, internal-state-preserving recoil channel (inelastic in total energy, although the internal state is preserved). An exciton at $\mathbf{K} = 0$ may absorb an LO phonon and transition to $1s$ at finite momentum $\mathbf{q}$. Energy conservation dictates

$$\hbar\omega_{\text{LO}} = \frac{\hbar^2 q^2}{2M_{\text{ex}}} \Rightarrow q_* = \frac{\sqrt{2M_{\text{ex}}\hbar\omega_{\text{LO}}}}{\hbar}. \quad (7)$$

The absorption rate is derived from Fermi's Golden Rule:

$$\Gamma_{\text{abs}} = \frac{2\pi}{\hbar} \frac{V}{(2\pi)^3} \int d^3q |G_{1s,1s}(\mathbf{q})|^2 N_{\text{LO}}(T) \delta\left(\frac{\hbar^2 q^2}{2M_{\text{ex}}} - \hbar\omega_{\text{LO}}\right). \quad (8)$$

Converting the delta function using $\delta(f(q)) = \delta(q - q^*)/|f'(q^*)|$ yields the Jacobian $|f'(q^*)|^{-1} = M_{\text{ex}}/(\hbar^2 q^*) = 1/(\hbar v_*(q^*))$, where v_* is the exciton group velocity at q_* . Writing $|M(q)|^2 = |\tilde{M}(q)|^2/V$ with $|\tilde{M}(q)|^2 = 2\pi e^2 \hbar\omega_{\text{LO}} (4\pi\epsilon_0 q^2)^{-1} (\epsilon_\infty^{-1} - \epsilon_s^{-1})$, the volume V from $\sum_{\mathbf{q}}$ cancels the $1/V$ exactly, so the rate is intensive. Performing the angular integration and evaluating the matrix elements on-shell gives the parameter-free recoil linewidth $\Gamma_{\text{recoil}}(T) = \gamma_{\text{LO}}^{\text{recoil}} N_{\text{LO}}(T)$, with

$$\gamma_{\text{LO}}^{\text{recoil}} = \frac{2\pi}{\hbar} \frac{q_*^2}{2\pi^2} \frac{|\tilde{M}(q_*)|^2 |\mathcal{F}_{1s,1s}(q_*)|^2}{\hbar v_*(q_*)}. \quad (9)$$

Invariant weight and status of the lattice-gauge split. The inclusive coupling weight $W(q) = \langle 1s | \hat{O}^\dagger \hat{O} | 1s \rangle = 2 - 2(1 + q^2 a_X^2/4)^{-2}$, with $\hat{O} = e^{if_e \mathbf{q} \cdot \mathbf{r}} - e^{-if_h \mathbf{q} \cdot \mathbf{r}}$, is a ground-state expectation value of the physical density operator and is therefore representation-independent; by the completeness relation it equals the sum of $|\mathcal{F}_{1s,s'}(q)|^2$ over all final excitonic states, of which the intraband term $|\mathcal{F}_{1s,1s}(q)|^2$ used throughout is one share. Disagreements

between bookkeepings are therefore disagreements about the intraband share of $W(q)$, never about $W(q)$ itself. For parabolic bands the center of mass separates exactly (Galilean invariance), the relative-coordinate vertex of Eq. (1) is the physical operator, and the continuum intraband coefficient $\frac{1}{2}\eta$ used throughout is exact. The lattice COM-gauge coefficient $\frac{3}{2}\eta$ [17] is obtained from a signed-minimum-image vertex which, for $q = 2\pi n/L$, $n \neq 0$, is

not the periodic lattice density operator; its replacement by the periodic vertex, and the resulting intraband share, are under audit in the revision of Ref. [17] (cluster ledger: OPEN). The rates in Table I are continuum-partition values, exact for parabolic bands; the inclusive weight is unaffected by this open item. Spontaneous emission is strictly forbidden as $E_X(\mathbf{q}) \geq E_X(0)$, ensuring $\Gamma \rightarrow 0$ as $T \rightarrow 0$.

When $\eta \rightarrow 0$, Eq. (3) forces $\gamma_{\text{LO}}^{\text{recoil}} = 0$. The linewidth is then dictated by the inelastic, internal-state-changing channel ($1s \rightarrow np$). The dipole form factor $\mathcal{F}_{1s,np}(q \rightarrow 0) = iqd_{np}$ is constructive and η -robust. The internal-state-changing channel is the constructive companion of Eq. (4): expanding the vertex to first order, $e^{if_e \mathbf{q} \cdot \mathbf{r}} - e^{-if_h \mathbf{q} \cdot \mathbf{r}} \approx i(f_e + f_h)\mathbf{q} \cdot \mathbf{r} = i\mathbf{q} \cdot \mathbf{r}$ since $f_e + f_h \equiv 1$, hence

$$\mathcal{F}_{1s,np}(q \rightarrow 0) = i\mathbf{q} \cdot \langle np | \mathbf{r} | 1s \rangle \equiv iqd_{np}, \quad (10)$$

independent of η to leading order. The corresponding resonant linewidth is

$$\gamma_{\text{LO}}^{\text{inel}} = \frac{2\pi}{\hbar} \sum_{n>1} \int \frac{d^3q}{(2\pi)^3} |G_{1s,np}(\mathbf{q})|^2 \delta_\gamma(\Delta E_{np} + E_{\text{recoil}} - \hbar\omega_{\text{LO}}), \quad (11)$$

where δ_γ accounts for the finite LO phonon lifetime. This channel is gated by the Rydberg detuning $\Delta E_{np} - \hbar\omega_{\text{LO}}$. Finally, if $\hbar\omega_{\text{LO}} > E_B$ (as in GaAs), the exciton may be ionized ($1s \rightarrow \text{continuum}$), adding a rate γ_{ion} .

III. UNIVERSAL SUPPRESSION OF THE PHONON CLOUD

Figure 1 shows the form factors: the internal-state-preserving channel (black) is suppressed by two orders of magnitude at small q relative to the constituent clouds, while the internal-state-changing channel (magenta) retains a finite dipolar strength. Consequently the Huang-Rhys integrand (Fig. 2) is confined to an intermediate- q window.

The integral $S_X \propto \int_0^\infty dq |\mathcal{F}_{1s,1s}(q)|^2$ evaluates in closed form. With $s = \sqrt{a} = f_e/2$, $t = \sqrt{b} = f_h/2$, the diagonal terms give $5\pi/(32s) + 5\pi/(32t)$ and the cross term

$$\int_0^\infty \frac{dq}{(1+aq^2)^2(1+bq^2)^2} = \frac{\pi [\sqrt{a}b^{3/2}(5a-b) + a^2(a-5b)]}{4\sqrt{a}(a-b)^3} \quad (12)$$

whence $S_{\text{ind}} \propto 5\pi(s+t)/(32st)$ and the ratio reduces exactly to Eq. (6). For MAPbI₃ parameters (Table I) we obtain $S_{\text{ind}} = 6.7$ but $S_X = 0.10$: the zero-phonon line retains $e^{-S_X} \approx 90\%$ of the spectral weight, versus $e^{-S_{\text{ind}}} \approx 0.1\%$ in the naive picture. Because both integrals scale as $1/a_X$, the ratio $R = S_X/S_{\text{ind}}$ is a one-parameter function of η alone *within the hydrogenic Fröhlich model*, given exactly by $\eta^2(6 - \eta^2)/5$ [Fig. 3].

IV. LINEWIDTHS: THE HIERARCHY OF CHANNELS

Table I and Fig. 5 establish the hierarchy of exciton-phonon scattering channels governed by $(\eta, q_*a_X, \Delta E_{np} - \hbar\omega_{\text{LO}})$.

Asymmetric regime (GaAs). With $\eta = -0.74$, parity protection fails at q_* ($\mathcal{F}(q_*) = -0.59$), yielding the parameter-free bound-state baseline $\gamma_{\text{LO}}^{\text{recoil}} = 2.2$ meV. Because $\hbar\omega_{\text{LO}} > E_B$, every bound-bound absorption channel is closed in GaAs ($\Delta E_{np} \leq E_B < \hbar\omega_{\text{LO}}$), so the LO-phonon absorption linewidth is entirely ionization broadening; its definitive treatment, with Coulomb-correlated continuum states, is Ref. [5], and the 8–14 meV bulk values compiled there are themselves determinations of that channel. The contribution of the present work is the recoil baseline above, a channel absent from frozen-COM treatments including Ref. [5]. A plane-wave (Born) evaluation of the ionization channel in the minimal model gives $\gamma_{\text{ion}}^{\text{PW}} \approx 110$ meV; this number is quoted only as a formal upper bound and is not used quantitatively. Note that GaAs ($q_*a_X = 8.35$) and MAPbI₃ (0.98) occupy dramatically different regions of (η, q_*a_X, Δ) : the observable is controlled by $|\tilde{M}(q_*)|^2 |\mathcal{F}(q_*)|^2 \rho_{\text{COM}}(q_*)$, not by α alone.

Symmetric regime (FAPbI₃). For exact mass symmetry ($\eta = 0$), Eq. (3) dictates $\gamma_{\text{LO}}^{\text{recoil}} = 0$ exactly. The internal-state-preserving channel is completely closed by parity.

Near-symmetric regime (MAPbI₃, CsPbBr₃). For MAPbI₃ ($\eta = -0.11$), the recoil rate is negligible (0.12 meV). The observed 27–40 meV linewidths therefore cannot be accounted for by COM recoil and require internal-state-changing and/or other channels; within the present model the $1s \rightarrow 2p$ resonance, Eq. (11), is the leading candidate. While a minimal model with a single dispersionless LO mode yields a small inelastic rate (~ 2 meV), perovskites possess a broad, highly dispersive soft-phonon spectrum that massively enhances the inelastic phase space. Full *ab initio* electron-phonon calculations are required to capture this absolute scale, but our model rigorously proves that the recoil channel is parity-blocked, forcing the broadening onto these inelastic channels.

The experimental column of Table I is organized by observable: Ref. 7 fits single-exciton homogeneous linewidths; Ref. 6 extracts carrier-Fröhlich strengths from film PL; Ref. 5 provides bulk exciton-LO comparisons.

V. DISCUSSION

α remains the natural measure of the constituent carrier-phonon coupling, but after projection onto the correlated exciton it is necessary rather than sufficient: the observables are fixed by $(\eta, q_*a_X, \Delta E_{np} - \hbar\omega_{\text{LO}})$. This yields design rules: engineer $\eta \rightarrow 0$ and detune the

Rydberg series from $\hbar\omega_{\text{LO}}$ to protect coherence. Limitations include the hydrogenic envelope, the dispersionless LO mode, and the estimated continuum floor; the selection rule itself is dimensionally invariant and extends to 2D systems [15]. Recent advances in *ab initio* many-body perturbation theory have enabled first-principles calculations of exciton–phonon matrix elements and finite-temperature relaxation times from the Bethe–Salpeter equation [19]. In such transition-basis formulations, the exciton–phonon matrix element is built from electron- and hole-phonon scattering amplitudes weighted by the exciton wave function. The analytical results derived here provide limiting cases that any such numerical calculation must satisfy: for an exactly mass-symmetric $1s$ exciton the internal-state-preserving Fröhlich recoil matrix element vanishes identically at all q , whereas the internal-state-changing $1s \rightarrow np$ dipole remains finite to leading order. Thus the present model is not a substitute for full *ab initio* phonon dispersions and transition-basis amplitudes, but supplies exact selection-rule benchmarks for those calculations.

VI. CONCLUSION

We have shown that restoring exact center-of-mass kinematics to the exciton–phonon problem exposes a universally open, internal-state-preserving absorption channel at the recoil momentum q_* . The exact finite- q identity $\mathcal{F}_{1s,1s}(q, \eta = 0) = 0$ proves that mass-symmetric excitons are perfectly protected from this recoil broadening. This establishes a strict hierarchy of scattering channels: monopolar COM-recoil coupling is suppressed by interference, while dipolar internal transitions and ionization channels survive. These results replace the Fröhlich constant α by the triplet $(\eta, q_* a_X, \Delta E_{np} - \hbar\omega_{\text{LO}})$ as the fundamental predictors of exciton–phonon scattering, providing rigorous design rules for coherent optoelectronic materials and establishing the necessary boundary conditions for future *ab initio* electron-phonon calculations.

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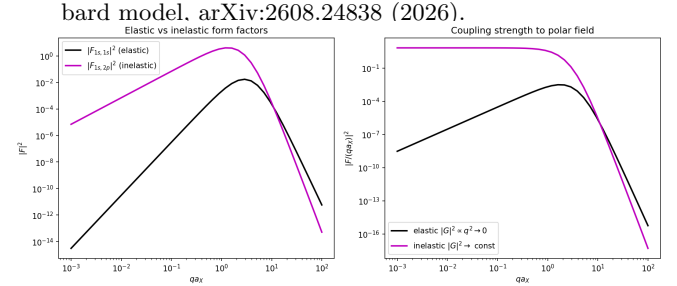


FIG. 1. (Color online) Internal-state-preserving (black) and internal-state-changing (magenta) excitonic form factors. Left: $|\mathcal{F}|^2$; right: coupling strength $|\mathcal{F}/(qa_X)|^2$ to the polar field. The internal-state-preserving vertex vanishes as q^2 (parity protection); the internal-state-changing vertex tends to a finite dipolar constant.

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TABLE I. Regime assignment and parameter-free rates. q_* : recoil momentum, Eq. (7); $\mathcal{F}(q_*)$: internal-state-preserving form factor at recoil; $\gamma_{\text{LO}}^{\text{recoil}}$: this work, Eq. (9); Experimental LO-related linewidth/coupling: experimental fits, organized by observable (footnotes).

	η	$q_* a_X$	$\mathcal{F}(q_*)$	$\gamma_{\text{LO}}^{\text{recoil}}$ (meV)	regime	$\gamma_{\text{LO}}^{\text{exp}}$ (meV)
GaAs	-0.74	8.35	-0.59	2.24	asymmetric: recoil + ionization	8–14 ^a
MAPbI ₃	-0.11	0.98	-0.04	0.12	near-symmetric: inelastic resonance	40 ± 2 ^b
FAPbI ₃	0.00	0.62	0.00	0.00	symmetric: inelastic resonance	27 ^c
CsPbBr ₃	-0.11	1.32	-0.07	0.65	intermediate	~ 70 ^d

^aBulk exciton–LO widths, Ref. 5; ^bcarrier–Fröhlich, Ref. 6; ^csingle-exciton linewidth, Ref. 7; ^dPL-broadening effective coupling, Ref. 18.

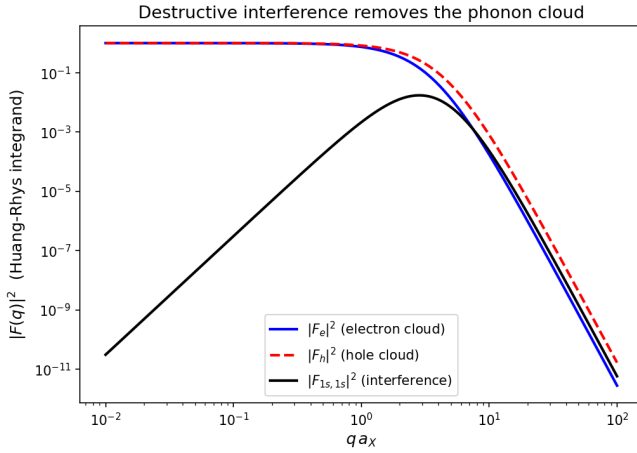


FIG. 2. (Color online) Huang–Rhys integrands. Destructive interference removes the small- q phonon cloud, confining the elastic dressing to intermediate q .

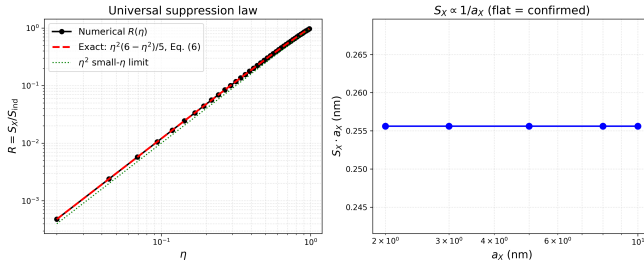


FIG. 3. (Color online) Universal suppression law. Left: $R(\eta) = S_X/S_{\text{ind}}$ (points) against the exact law $\eta^2(6 - \eta^2)/5$, Eq. (6) (dashed). Right: exact $1/a_X$ scaling of S_X .

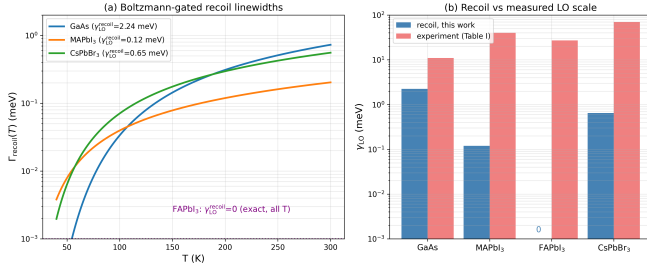


FIG. 4. (Color online) (a) Temperature dependence of the internal-state-preserving recoil linewidths $\Gamma_{\text{recoil}}(T) = \gamma_{\text{LO}}^{\text{recoil}} N_{\text{LO}}(T)$ using the parameter-free baselines of Table I; for FAPbI₃ the rate is exactly zero at all T by Eq. (3). (b) Recoil coefficients (blue) against the measured LO-related scales (red, Table I). The recoil channel supplies the GaAs bound-state baseline but lies two to three orders of magnitude below the perovskite benchmarks, which therefore require internal-state-changing channels (not shown).

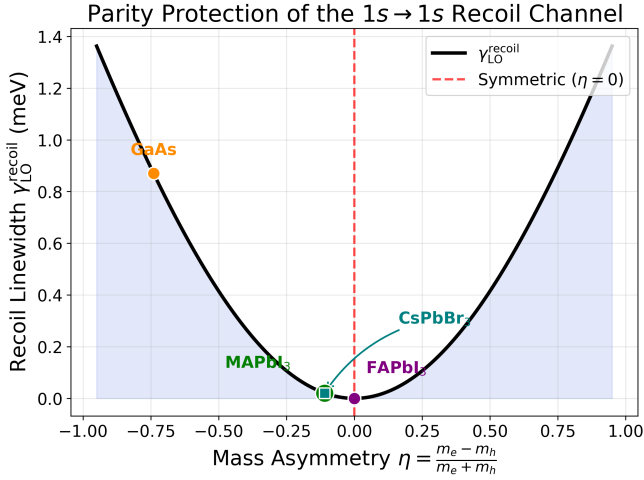


FIG. 5. (Color online) Parity protection of the $1s \rightarrow 1s$ COM recoil channel. The parameter-free recoil linewidth, Eq. (9), vanishes exactly at $\eta = 0$ (mass symmetry). Mass-symmetric perovskites sit at the protected minimum, forcing their linewidth onto the inelastic $1s \rightarrow np$ channel; mass-asymmetric GaAs sits on the unprotected branch.